

MACHINE LEARNING PROJECT

Credit Score Prediction

Data Analysis, Feature Engineering, and Predictive Modeling for Bank Customers Credit Score Data

PREPARED BY

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14/2/2026



Agenda

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Business Objective

Evaluate people to measure the **risk of lending money** to someone.



Credit Score

A numerical rating of a person's credit worthiness, based on their credit history.



300 – 850

credit score range

Dataset Description

Data columns



Delay from due date

Number of delayed days from the due date



Type of loan

e.g. Personal loan, Auto loan



Num of bank accounts

Bank accounts held by the customer



Num of credit cards

Credit cards held by the customer



Num of loan

Number of loans taken



Credit history age

Age of the credit account



Payment of minimum amount

Did he pay the minimum limit or more?



Interest rate

Interest rate on the loans

Unimportant columns

Name

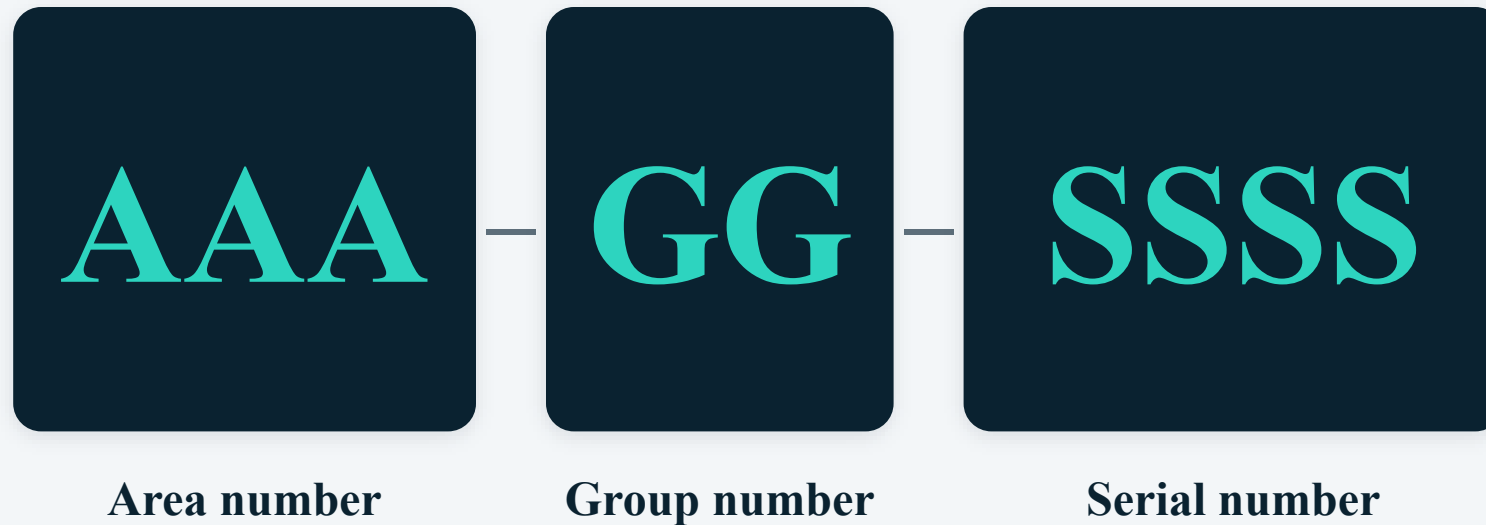
ID

Customer ID

Dataset Description

Data columns · SSN (Social Security Number)

SSN = AAA - GG - SSSS



Good to know

After 25 June 2011, the Social Security Administration changed the system to be randomized.

Target Distribution

Credit Score

CREDIT SCORE BANDS

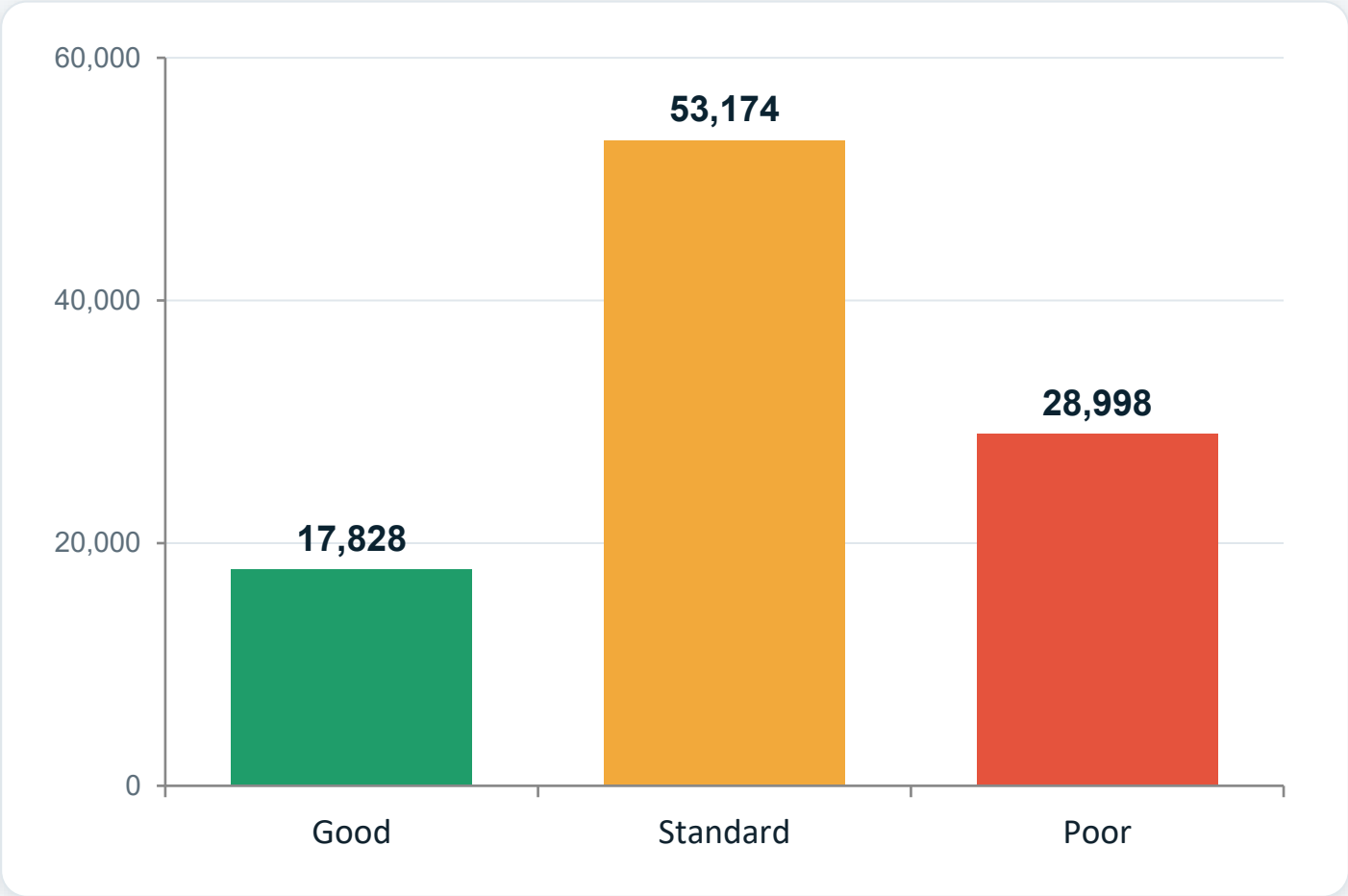
800 – 850	Excellent
740 – 799	Very Good
670 – 739	Good
580 – 669	Standard / Fair
300 – 579	Poor

TARGET CLASSES

Good	The customer usually pays debts on time
Standard	Average payment behavior, sometimes late
Poor	Frequent late payments or credit problems

Target Distribution

Credit Score count



● Good

17,828

● Standard

53,174

● Poor

28,998

● Missing (unlabeled)

50,000

Unbalanced distribution

Invalid Values

50,000

unlabeled rows → dropped from the data frame

Column	Invalid value	New value
Occupation	_____	undetermined occupation
Credit mix	—	undetermined credit mix
Payment behavior	!@9#%8	Undefined behavior

Invalid Values

Column's type correction



Outstanding debt

Age

Annual income

Num of delayed payment

Changed credit limit

Num of loan

Monthly balance

Amount invested monthly

Outliers

Checking the outliers of Age column

2.781%

outliers percentage

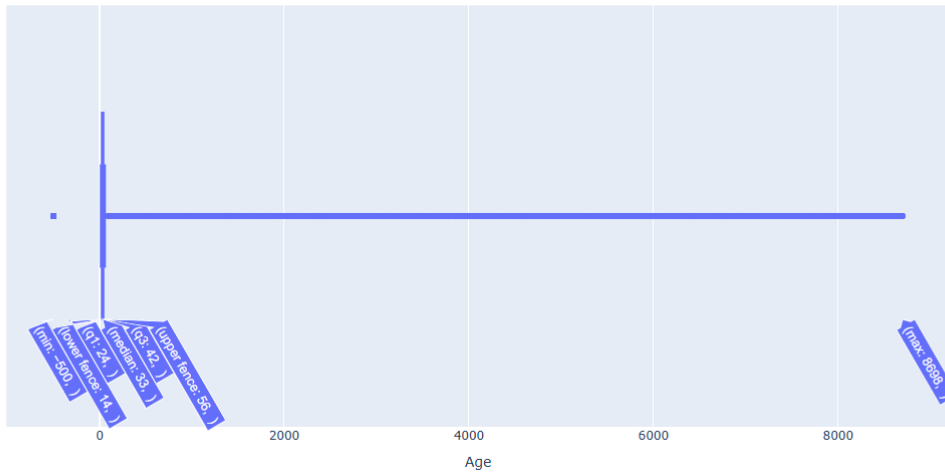
Age < 21 years old

replaced by median = 33

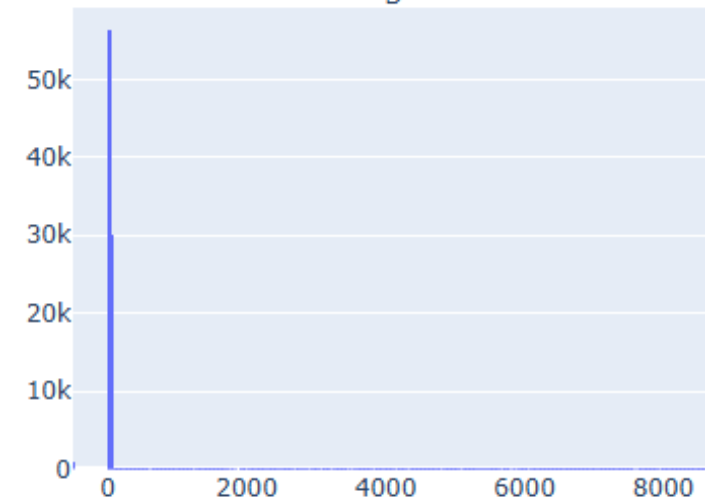
Age > 69 years old

replaced by median = 33

Age Boxplot



Age



Outliers

Checking the outliers of Outstanding Debt column

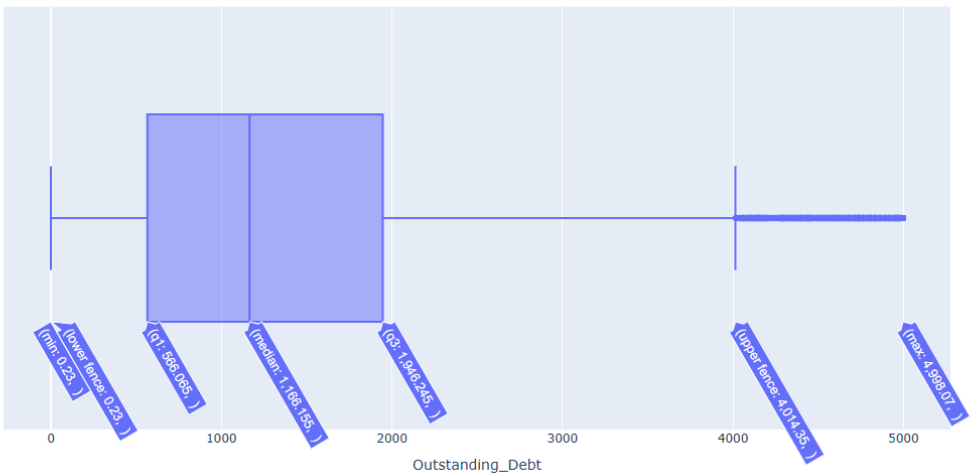
5.272%

outliers percentage

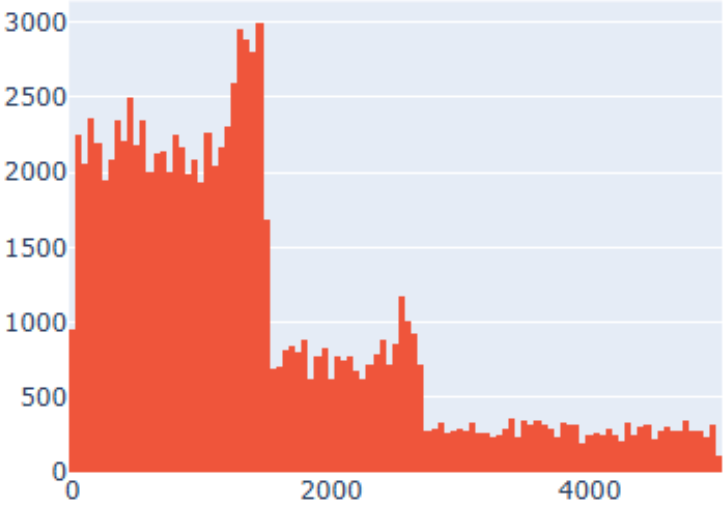
Max outlier: 5,000

acceptable deviation

Outstanding_Debt Boxplot

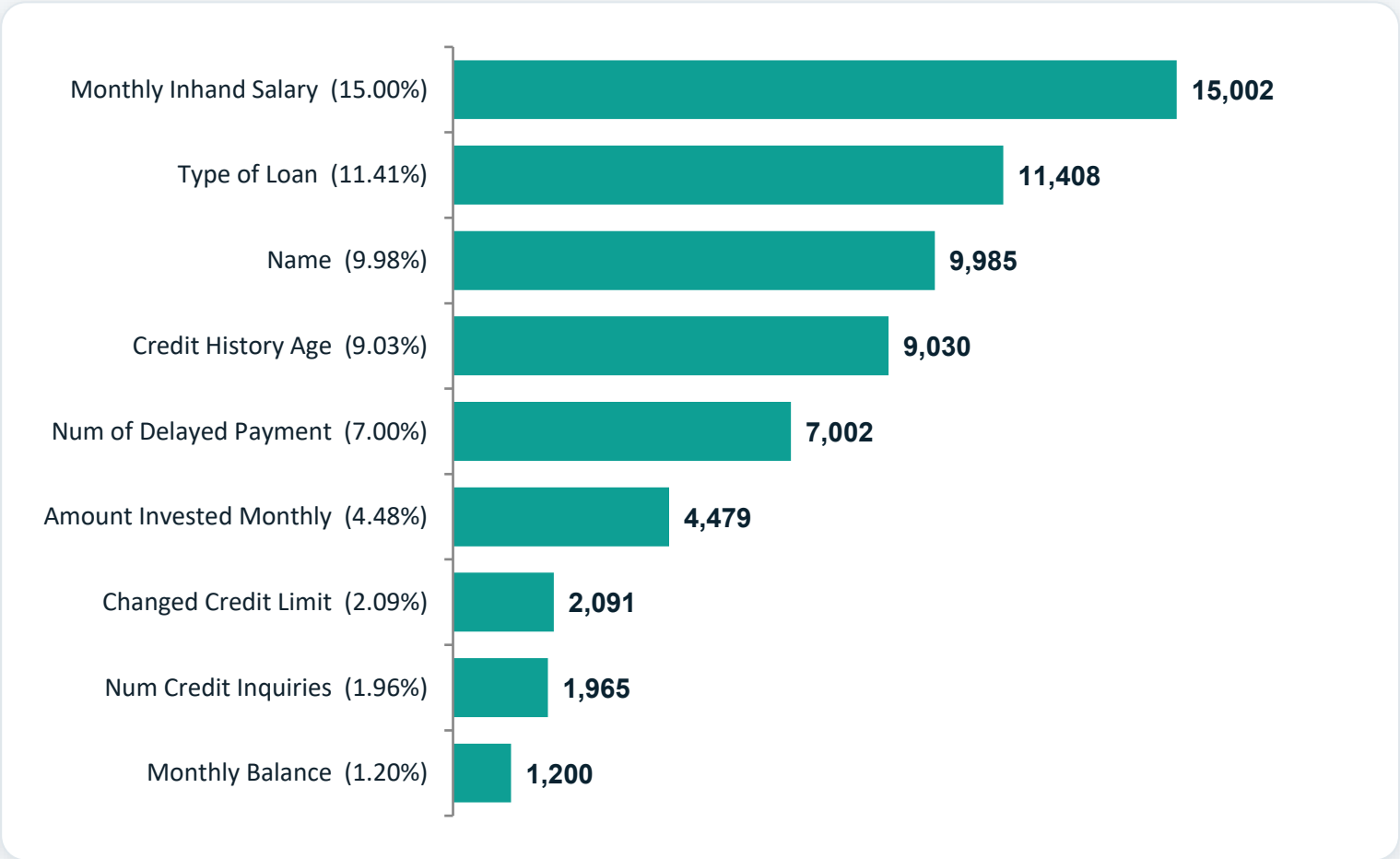


Outstanding_Debt



Missing Values

Number of missing values per column



62,162
total missing values

Numerical
filled with the median

Categorical
filled with the mode

Feature Engineering

Four financial ratios created from the raw columns

1 EMI to Income Ratio

$$\frac{\text{total EMI per month}}{\text{monthly inhand salary}} \times 100\%$$

Ratio between the monthly EMI and the monthly in-hand salary.

Risk factor: 40%

2 Debt to Income

$$\frac{\text{outstanding debt}}{\text{annual income}} \times 100\%$$

< 30% good

30–50% not bad

> 50% risky

3 Invested to Income Ratio

$$\frac{\text{amount invested monthly}}{\text{monthly inhand salary}} \times 100\%$$

Ratio between the monthly invested amount and the monthly in-hand salary.

4 Balance to Income Ratio

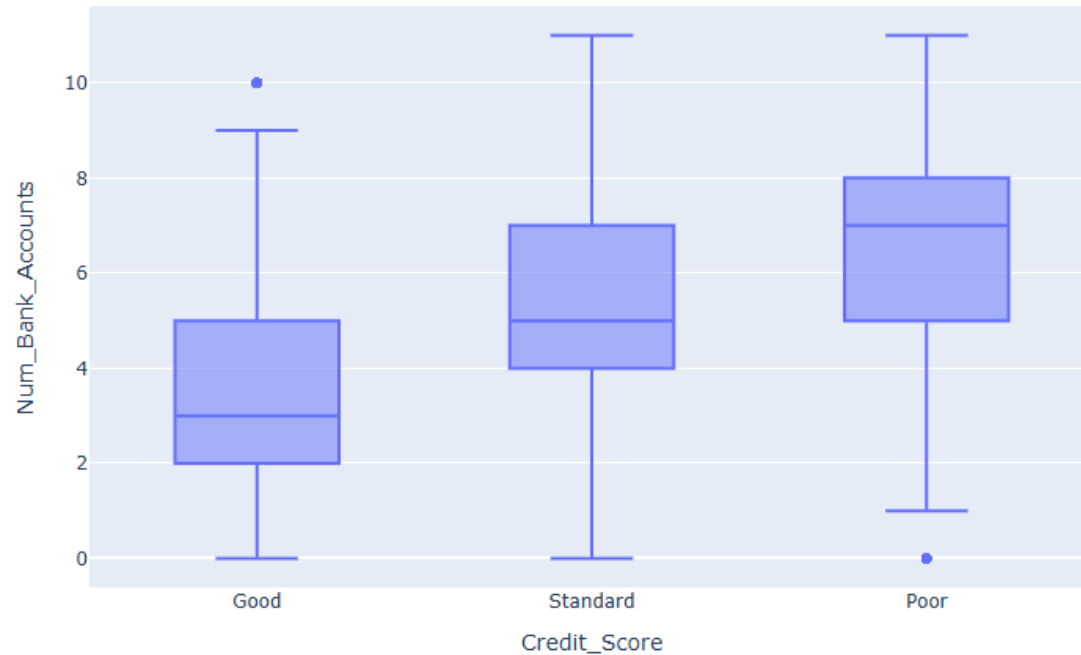
$$\frac{\text{monthly balance}}{\text{monthly inhand salary}} \times 100\%$$

Ratio between the monthly balance and the monthly in-hand salary.

Exploratory Data Analysis (EDA)

Num_Bank_Accounts by Credit_Score

Num_Bank_Accounts by Credit_Score



MEDIAN BANK ACCOUNTS

● Good ≈ 3

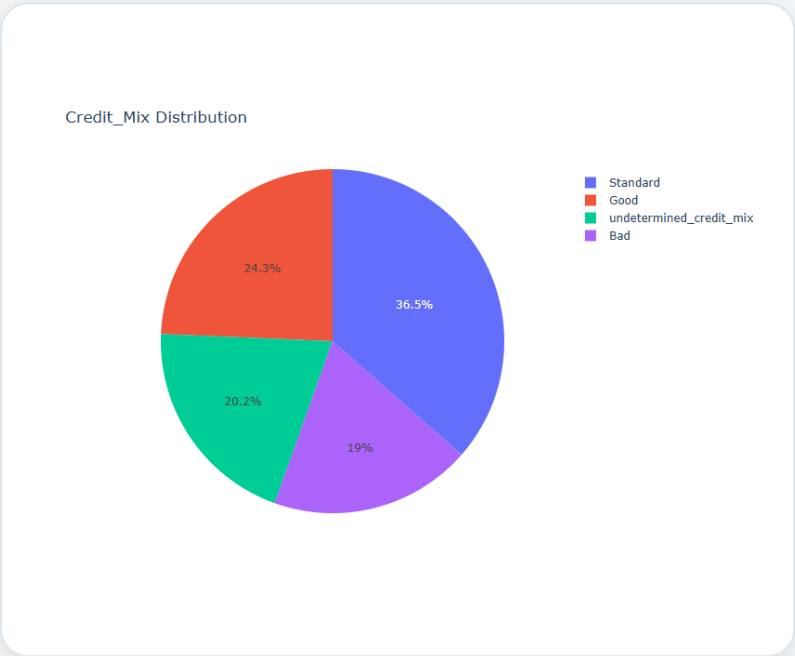
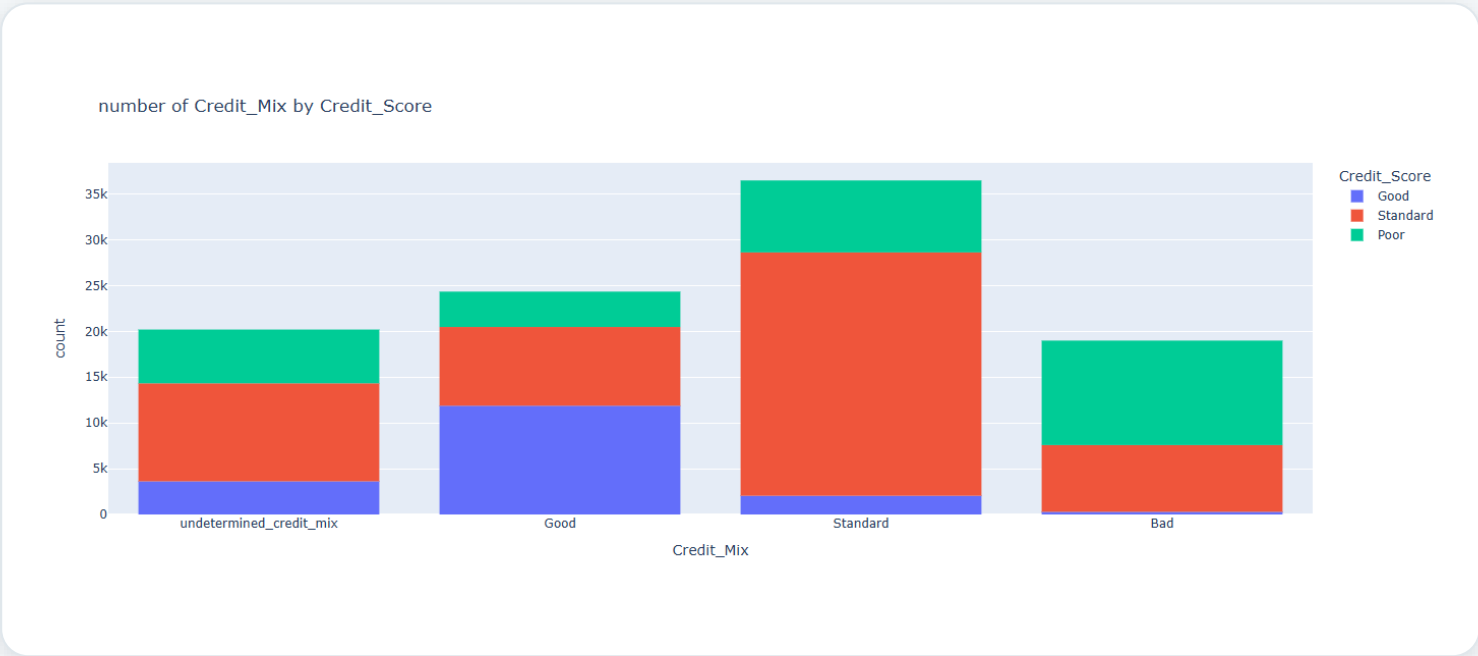
● Standard ≈ 5

● Poor ≈ 7

More bank accounts tend to go with a worse credit score.

Exploratory Data Analysis (EDA)

Credit_Mix distribution and its relation to Credit_Score



Good credit mix is mostly linked to Good scores



Bad credit mix is mostly linked to Poor scores

Exploratory Data Analysis (EDA)

Selected features

NUMERICAL

1 Age

2 Monthly balance

3 Credit history age

4 Num bank accounts

5 Num credit inquiries

6 Num of delayed payment

7 Delay from due date

8 Num of loan

9 Interest rate

10 Num credit card

11 Outstanding debt

12 Monthly inhand salary

13 Annual income

14 Debt to income

CATEGORICAL

1 Type of loan

2 Credit mix

3 Payment behaviour

4 Payment of min amount

Encoding and Scaling

Encoding of the categorical features

1 Type of loan

Student loan

Personal loan

Debt consolidation loan

Home equity loan

Payday loan

Credit builder loan

Not specified

Auto loan

Mortgage loan

2 Credit mix

0 Undetermined credit mix

1 Good

2 Standard

3 Bad

3 Payment behaviour

0 Undefined behaviour

1 High spent large value payment

2 High spent medium value payment

3 High spent small value payment

4 Low spent large value payment

5 Low spent medium value payment

6 Low spent small value payment

4 Payment of min amount

0 NM "not mentioned"

1 No

2 Yes

Encoding and Scaling

Scaling



**Normalize the
numerical features**

14

numerical features

Age

Monthly balance

Credit history age

Num bank accounts

Num credit inquiries

Num of delayed payment

Delay from due date

Num of loan

Interest rate

Num credit card

Outstanding debt

Monthly inhand salary

Annual income

Debt to income

Modeling and Evaluation



Data balancing

Under-sampling



Train / test split

80% training · 20% testing



Models implemented

Random Forest and Logistic Regression

ACCURACY BY MODEL



Grid Random Forest parameters

150

n-estimators

13

max-depth

5

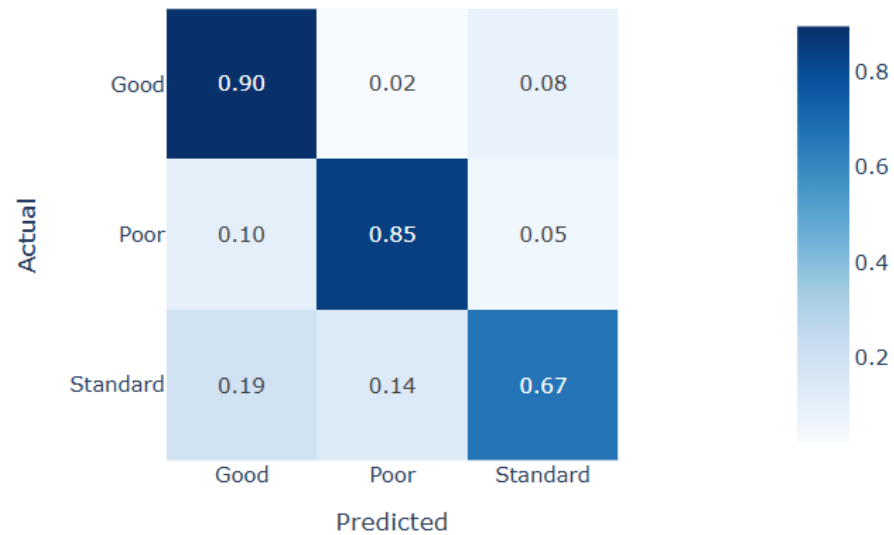
stratified K-fold
splits

Modeling and Evaluation

Evaluation using confusion matrix – Random Forest model

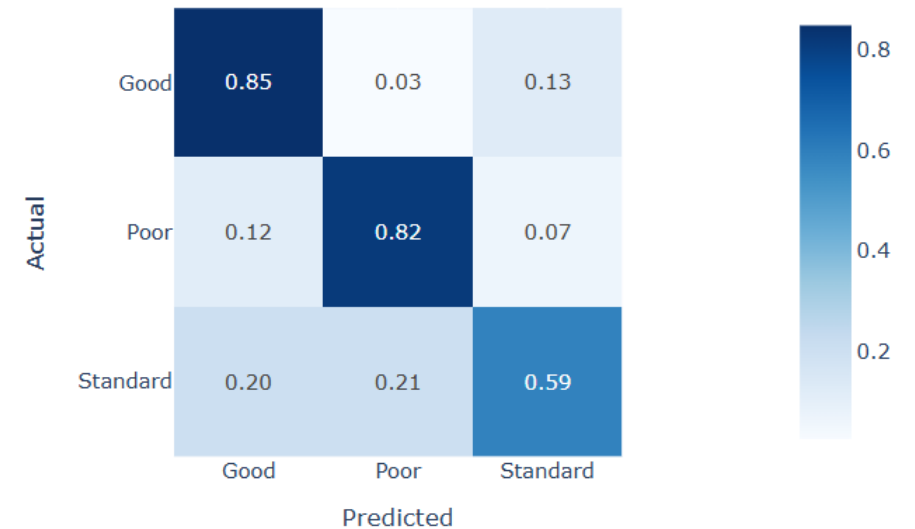
Training set prediction

Accuracy **80%**



Testing set prediction

Accuracy **75%**



Feature Importance

Final selected features after dropping features with importance less than 0.04

9

features kept

Features with importance < 0.04 were dropped.

1

Credit history age

2

Outstanding debt

3

Num credit inquiries

4

Delay from due date

5

Interest rate

6

Num bank accounts

7

Debt to income

8

Payment of minimum amount

9

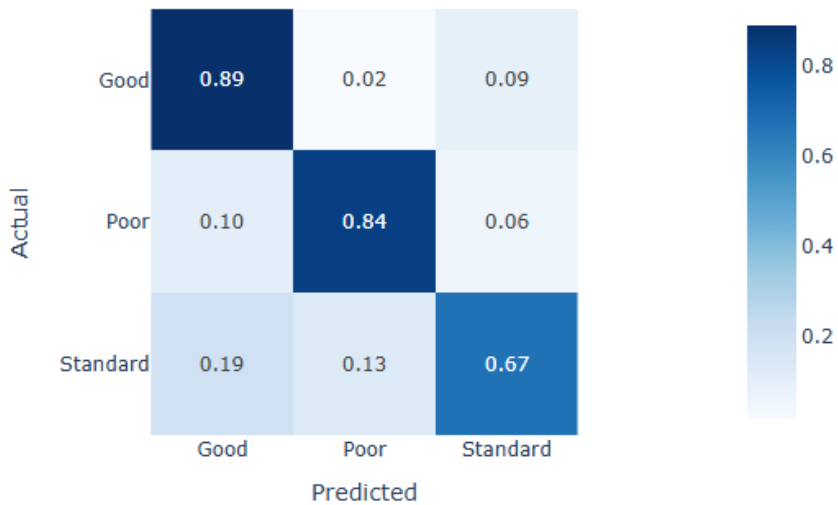
Credit mix

Feature Importance

Evaluation using confusion matrix – Random Forest model with the 9 selected features

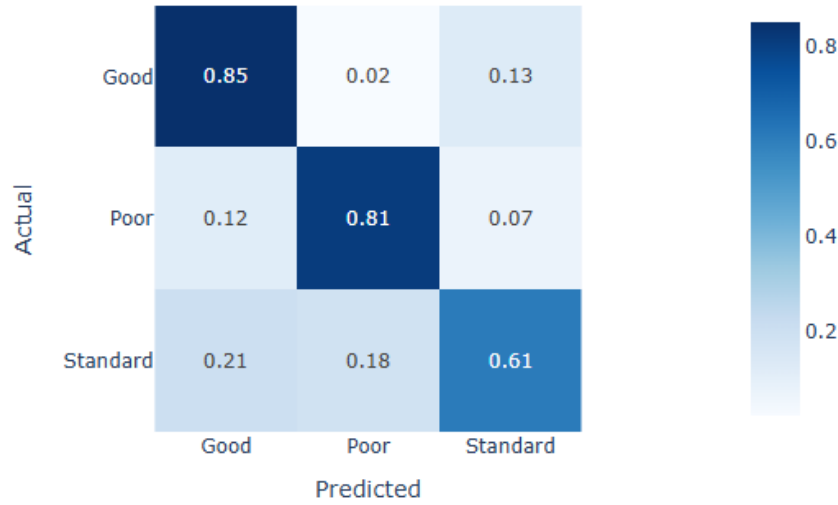
Training set prediction

Accuracy **80%**



Testing set prediction

Accuracy **76%**



Test accuracy improved from 75% to 76% using only the 9 selected features

Summary



Data preparation

100,000 labeled rows. Invalid values fixed, types corrected, outliers and missing values handled (median / mode).



Feature engineering

Four income ratios: EMI, debt, invested and balance to income, plus encoding and scaling.



Modeling

Random Forest (150 trees, depth 13, 5-fold) beat Logistic Regression: 76% vs 66%.



Top drivers

Credit history age, outstanding debt, credit inquiries, delay from due date and interest rate.



Final model

9 selected features reach 80% training and 76% testing accuracy.

76%

test accuracy of the final Random Forest model

Live dashboard

[Open the Streamlit dashboard](https://creditscore-dashboard.streamlit.app/)



<https://creditscore-dashboard.streamlit.app/>

Thank You

Questions & Discussion

[Explore the Interactive dashboard](#)



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